Density altitude and lift

Edition Oct 2026

Three steps

1 · Pressure altitude
PA = elevation + (29.92 − altimeter) × 1,000

2 · Standard temperature
ISA = 15 °C − 2 °C per 1,000 ft of PA

3 · Density altitude
Chart: up from OAT to the PA line, then across. Or a flight computer (E6B).

Worked example

Field elevation 1,400 ft, altimeter 29.82, OAT 30 °C.

Pressure altitude	1,400 + (29.92 − 29.82) × 1,000 = 1,400 + 100 = **1,500 ft**
Standard temperature	15 − 2 × 1.5 = **12 °C**
Density altitude	30 °C is 18° above standard. Chart: up from 30 °C to the 1,500 ft line, then across: **about 3,600 ft**

About 3,600 ft
The balloon performs as if the field were about 2,200 ft higher.

What it does to lift

- Buoyancy equals the weight of the air the envelope displaces. Heating the air inside makes it less dense, so the balloon rises when it weighs less than the air it pushes aside.
- High elevation, low pressure, high temperature and high humidity all mean thinner air and less lift at the same envelope temperature: more heat for the same load, more fuel, less margin under the envelope's limit.
- Rough rule: 1,000 cubic feet of heated air lifts about 20 lb.
- Burner performance drops about 4% per 1,000 ft.
- Maximum gross weight is in your flight manual; its performance chart reads lift from ambient temperature, envelope temperature and pressure altitude.

Worksheet

Step	Flight 1	Flight 2
Field elevation (ft)		
Altimeter setting (inHg)		
(29.92 − altimeter) × 1,000		
Pressure altitude		
ISA = 15 − 2 × (PA ÷ 1,000)		
OAT (°C)		
OAT − ISA (+ warmer, − colder)		
Density altitude (chart or E6B)		

Check your method on the FAA's own example: pressure altitude 5,048 ft at 30 °C is a density altitude of 7,855 ft (PHAK p. 8-7). The Aviation Weather Handbook computes density altitude with a flight computer (p. 8-15).

Sources: FAA-H-8083-25C Ch. 4 (standard atmosphere, pressure and density altitude), FAA-H-8083-25C p. 8-7 (worked density altitude example), FAA-H-8083-11B Ch. 2 (buoyancy), FAA-H-8083-11B Ch. 3 (performance planning), FAA-H-8083-28B p. 8-15 (density altitude: example, flight computer). Original card, not an FAA document. Study aid, not for navigation.